

1.4 Summary

In this chapter, we introduced functional programming and explained exactly what FP is and why you might use it. Though the full benefits of the functional style will become more clear over the course of this book, we illustrated some of the benefits of FP using a simple example. We also discussed referential transparency and the substitution model and talked about how FP enables simpler reasoning about programs and greater modularity.

In this book, you'll learn the concepts and principles of FP as they apply to every level of programming, starting from the simplest of tasks and building on that foundation. In subsequent chapters, we'll cover some of the fundamentals—how do we write loops in FP? Or implement data structures? How do we deal with errors and exceptions? We need to learn how to do these things and get comfortable with the low-level idioms of FP. We'll build on this understanding when we explore functional design techniques in parts 2, 3, and 4.